



小车运行机构设计计算书

Calculation Sheet for Design of Trolley Operation Mechanism

1 小车运行机构

1. Travelling Mechanism of Hoist Trolley

机构级别:	M8	
Mechanism grade:		
	T6	L3
小车重量:	TL= 4.9	t
Hoist trolley weight:		
小车基距:	Bx= 1800	mm
Hoist trolley base:		
小车轨距:	Ly= 1800	mm
wheel thread of hoist trolley:		
小车重心高度 (相对轨面):		(满载, 相对于轨顶面)
Gravity center height of hoist trolley	$h_{gt1}= 1037$	mm
(relative track):		(Full load, relative to the track top surface)
	$h_{gt2}= 351$	mm
		(空载)
		(No Load)
小车满载运行速度:		
Travelling speed of hoist trolley at full	$v_{y1}= 25$	m/min
load:		
	0.42	m/s
小车空载运行速度:		
Travelling speed of hoist trolley at no load:	$v_{y2}= 25$	m/min
	0.42	m/s
许用制动时间:		
Allowable braking time:	$[t_z]= 4.0$	s
许用起动时间:		
Allowable starting time:	$[t_q]= 4.0$	s
许用起、制动加速度:		
Allowable starting and braking	$[a_z]= 0.10$	m/s ²
acceleration:		
小车车轮数:		
Number of hoist trolley wheels:	$n_h= 4$	

2. 车轮和轨道

2. Wheel and Trolley Rail

(1) 选型

(1) Model Selection

选择车轮直径:	$D_{yt}= 200$	mm
Diameter of selected wheel:		
车轮内孔直径:	$d_{it}= 70$	mm

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Wheel inner hole diameter:

车轮宽度: $B=80$ mm
Wheel width:

车轮型号: **ZQCL200B**
Wheel model: **ZQCL200B**

车轮材料: **42CrMo**
Wheel material: **42CrMo**
 $\sigma_b=1080$ MPa

单个车轮装配重量:
Assembly weight of single wheel: $m_{it}=50$ kg

车轮轴材料: **42CrMo**
Material of wheel axle: **42CrMo**

车轮轴直径: $d_{it}=70$ mm
Diameter of wheel axle:

小车轨道型号: **方钢**
Trolley rail model of hoist trolley: **Square Steel**

轨道尺寸: $h1=30$ mm
Trolley rail size:
 $b1=50$ mm
 $k=50$ mm
 $r1=0$ mm

轨道材料: **C45 钢**
Trolley rail material: **C45 Steel**

$\sigma_b=560$ MPa
(强度极限应大于 900MPa)
(The strength limit shall be greater than 900MPa)

(2) 踏面疲劳强度校核 (2) Tread Fatigue Strength Check

小车自重引起的轮压:
Wheel load caused by the hoist trolley's self-weight: $P_{Lt11}=12.02$ kN

(满载) $\Phi 4$ 1.475 17.73
(Full Load)

$P_{Lt12}=12.02$ kN
(空载) $\Phi 4$ 1.475 17.73
(No Load)

起升载荷引起的轮压: $P_{Lt11}=24.53$ kN
Wheel load caused by moving load:

(满载) $\Phi 4$ 1.475 36.17
(Full Load)

$P_{Lt12}=9.32$ kN
(空载) $\Phi 4$ 1.475 13.75
(No Load)

由惯性载荷引起的轮压: $P_{Lt31}=0.45$ kN
Wheel load caused by inertial load:

(满载) $\Phi 2, f4$ 7.18
(Full Load)

$P_{Lt32}=0.09$ kN
(空载) $\Phi 2, f4$ 0.12
(No Load)

正常工作状态下的最小轮压:
Minimum wheel load under the normal working status: $P_{Ltmin}=21.25$ kN

正常工作状态下的最大轮压: $P_{Ltmax}=36.99$ kN 61.079

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Maximum wheel load under the normal
working status:

平均轮压: $P_{Lmean} = 31.74$ kN
Average wheel load:

$$\left(P_{Lmean} = \frac{P_{Lmin} + 2P_{Lmax}}{3} \right)$$

与材料有关的许用线接触应力常数:

Allowable line contact stress constant

related with material: $k_1 = 7.20$

N/mm²

($\sigma_b > 510$)

($\sigma_b > 510$)

车轮与轨道有效接触宽度: $l =$

50 mm

Effective contact width between the wheel
and trolley rail:

车轮转速:

Wheel rotation speed: $n_{yt} = 39.79$

r/min

转速系数:

Speed coefficient: $C_1 = 0.97$

(根据车轮直径和运行速度)

(Based on the wheel diameter and travelling speed)

工作级别系数:

Working grade coefficient: $C_2 = 0.80$

(机构级别 M8)

(Mechanism grade: M8)

校核公式:

Check formula: $P_{Lmean} \leq k_1 D_{yt} l C_1 C_2$

$$k_1 D_{yt} l C_1 C_2 = 55.87 \text{ kN}$$

车轮满足疲劳强度要求!

The wheel meets the requirements for fatigue strength!

静强度校核:

Static strength check: $P_{Lmax} \leq 1.9 k_1 D_{yt} l$

$$1.9 k_1 D_{yt} l = 136.80 \text{ kN}$$

车轮满足静强度要求!

The wheel meets the requirements for static strength!

3. 轴承

3. Bearing

调心滚子轴承:

Self-aligning rolling bearing: 22212 CC/W33

重量:

Weight: 2.68 kg

轴承支承距离:

Bearing support distance: $L_z = 184$

mm

(车轮中心到轴承中心的距离)

(Distance from wheel center to bearing center)

轴承外径:

Bearing outer diameter: $D = 110$

mm

轴的直径:

Shaft diameter: $d_z = 60$

mm

倒角:

Chamfer: $r = 1.5$

mm

$$L_{10h} = \frac{10^6}{60 n_{yt}} \left(\frac{f_t \times C_r}{P} \right)^{\epsilon}, P = f_d (X F_r + Y F_a)$$

$$C_{r0} = 155 \text{ kN}$$

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额定寿命: Rated life:	$[L_{10h}] = 12500$	小时 Hour	(与机构的使用等级有关) T6 (It is related to the use grade of mechanism) T6
温度系数: Temperature coefficient:	$f_T = 1.00$		(工作温度 $< 120^\circ\text{C}$) (Operating temperature: $< 120^\circ\text{C}$)
寿命系数: Life coefficient:	$\varepsilon = 0.24$		(滚子轴承) (Rolling bearing)
冲击载荷系数: Impact load efficient:	$f_d = 1.20$		(中等冲击力) (Medium impact force)
轴向载荷系数: Axial load coefficient:	$\gamma = 0.10$		
实际轴向载荷: Actual axial load:	$F_{a1} = 0.00$	kN	
	$F_{a2} = 3.70$	kN	$F_a = \gamma P_L$
实际径向载荷: Actual radial load:	$F_{r1} = 20.50$	kN	$\left(F_r = \frac{P_L}{2} \pm \frac{F_a}{L_z} \cdot \frac{D_{yc}}{2} \right)$
	$F_{r2} = 16.48$	kN	
	$F_a/F_r = 0.00$	0.22439	
判断系数: Judgment coefficient:	$e = 0.30$		
径向动载系数: Radial dynamic load coefficient:	$X = 1.00$	0.67	
轴向动载系数: Axial dynamic load coefficient:	$Y = 2.80$	4.10	
当量动载荷: Equivalent dynamic load:	$P_1 = 24.61$	kN	
	$P_2 = 31.45$	kN	
	$P = 26.89$	kN	
额定动载荷: Rated dynamic load:	$C_r = 122$	kN	
实际轴承寿命: Actual bearing life:	$L_{10h} = 64778$	小时 Hour	轴承满足寿命要求! The bearing meets the life requirements!

4. 三合一减速机

4. Three-in-one Reducer

(1) 载荷计算

(1) Load Calculation

自重载荷: Self-weight load:	$P_{Gt1} = 146.2$	kN	(满载) (Full Load)
	$P_{Gt2} = 85.3$	kN	(空载) (No Load)
小车水平惯性载荷: Horizontal inertial load of hoist trolley:	$P_{At1} = 1.6$	kN	(满载) (Full Load)
	$P_{At2} = 0.9$	kN	(空载) (No Load)
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车轮轴承摩擦系数: Wheel bearing friction coefficient:	$\mu_t = 0.015$	(No Load) (滚子轴承) (Rolling bearing)
滚动摩擦系数: Rolling friction coefficient:	$k_r = 0.30$	(轨道为钢质、平顶) (The trolley rail is made of steel with flat top)
附加摩擦阻力系数: Additional friction resistance coefficient:	$C_f = 1.30$	(有轮缘有水平滚轮, 分别驱动) (It is designed with wheel rim and horizontal roller, driven respectively)
运行摩擦阻力: Travelling friction resistance:	$P_m = mg \frac{\mu d + 2k}{D_{yt}} C_f$	
	$P_{mt1} = 1.60 \quad \text{kN}$	(满载) (Full Load)
	$P_{mt2} = 0.93 \quad \text{kN}$	(空载) (No Load)
运行坡度阻力: Travelling grade resistance:	$P_p = mg \cdot \sin \alpha = mg \cdot m_p$	
坡道阻力系数: Slope resistance coefficient:	$m_p = 0.002$	
	$P_{pt1} = 0.30 \quad \text{kN}$	(满载) (Full Load)
	$P_{pt2} = 0.17 \quad \text{kN}$	(空载) (No Load)
运行风阻力: Travelling wind resistance:	$P_{wl} = Cq_w A_w$	
风力系数: Wind coefficient:	$C = 1.10$	
	$P_{wt1} = 0.0 \quad \text{kN}$	(满载) (Full Load)
	$P_{wt2} = 0.0 \quad \text{kN}$	(空载) (No Load)
小车运行静阻力: Static resistance of hoist trolley travelling:	$P_j = P_m + P_p + P_{wl}$	
	$P_{jt1} = 1.9 \quad \text{kN}$	(满载) (Full Load)
	$P_{jt2} = 1.1 \quad \text{kN}$	(空载) (No Load)

(2) 电动机选型

(2) Model Selection of Electric Motor

电动机台数: Quantity of electric motor:	$m = 2$	(分别驱动) (It is driven respectively)
运行机构总效率: Total efficiency of travelling mechanism:	$\eta_{tz} = 0.95$	
机构静功率: Mechanism's static power:		

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$$P_N = \frac{P_{jt} \cdot v_{yt}}{1000 \eta_{tz} \cdot m} = 0.42 \quad \text{kW}$$

考虑 1.4 倍增大系数
Considering the 1.4-fold increasing coefficient 0.58 kW

所选电动机的参数如下:
The parameters of the selected motor are as follows:

型号: 诺德 (三合一平行轴减速机) 减速机 P272
Model: **NORD (three-in-one parallel shaft reducer)** Reducer P272
SK1382.1AXZNBGH-80L/4

额定功率: $P_n = 0.75$ kW
Rated power:
温度系数: $f_T = 0.77$ (50°C)
Temperature coefficient: (50°C)

起动系数: $f_H = 0.98$
Starting coefficient:

额定转速: $n_n = 1415$ r/min
Rated speed:

机构的理论传动比:
Mechanism's the theoretical transmission ratio: $i_0 = 35.56$ ($i_0 = n_n / n_{yt}$)

实际传动比: $i = 38.67$
Actual transmission ratio:

电动机输出转矩:
Output torque of electric motor: $M_2 = 5.07$ N·m

平均起动转矩倍数: $\lambda_{as} = 1.60$
Average starting torque multiple: **平均起动转矩不可过大, 否则造成空载加速度过大!**
The average starting torque shall not be too large otherwise it may lead to excessive no-load acceleration!

最大起动转矩倍数: $\lambda_m = 2.4$
Maximum starting torque multiple:

转动惯量: $J_1 = 0.00261$ kg·m²
Rotational inertia:

电动机选 满足要求!
Motor Selection It meets the requirements!

(3) 实际电动机转速 (3) Actual Speed of Electric Motor

转速校核公式: $n_N = \frac{v_{yt} i}{\pi D_{yt}}$
Speed check formula:

电动机实际转速: $n_{yc} = 1539$ m/min (满载)
Actual speed of electric motor: (Full Load)

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5. 高速轴制动器

5. High-speed Shaft Brake

$$M_z = \frac{1}{m} \left\{ \frac{(P_{wt} + P_p - P_m) D_{yt} \cdot \eta}{2i} + \frac{n \sum J}{9.55 t_z} \right\}$$

电机轴上转动件的转动惯量:

Rotational inertia of rotating parts on the motor shaft: $J_2 = 0.000 \text{ kg} \cdot \text{m}^2$

折算到电机轴上的总转动惯量:

Total rotational inertia converting to motor shaft: $\sum J' = 0.101 \text{ kg} \cdot \text{m}^2$ $\sum J' = 1.15m(J_1 + J_2) + \frac{(Q + TL)D_0^2 \eta}{4i^2}$

制动器的制动力矩: $M_z = 0.27 \text{ N} \cdot \text{m}$
Braking torque of brake:

最大制动力矩: $M_{z\max} = 10 \text{ N} \cdot \text{m}$
Maximum braking torque:

允许制动力矩: $[M_z] = 0.5 \text{ N} \cdot \text{m}$
Allowable braking torque:

电动机自带 BR55IW 系列制动器

The electric motor is self-provided with the BR55IW series brake

制动力矩满足要求!

The braking torque meets the requirements!

重量: $m_{31} = 91 \text{ kg}$
Weight:

个数: $m_z = 2 \text{ Unit}$
Qty.:

制动器选择满足要求!

The brake selection meets the requirements!

6. 机构起动时间和加速度的验算

6. Calculation of Mechanism Starting Time and Acceleration

(1) 满载起动时间和起动平均加速度验算

(1) Check Calculation for Starting Time and Average Starting Acceleration at Full Load

① 起动时间

① Starting Time

$$t_q = \frac{n \sum J}{9.55 (m M_{dq} - M_{dj})}$$

电动机起动力矩: $M_{dq} = 14.9 \text{ N} \cdot \text{m}$
Starting torque of electric motor:

电动机静阻力矩: $M_{dj} = 5.2 \text{ N} \cdot \text{m}$
Static torque of electric motor: $\left(M_{dj} = \frac{P_{jt} D_{yt}}{2000 i \eta} \right)$

折算到电机轴上的总转动惯量:

Total rotational inertia converting to motor shaft: $\sum J = 0.111 \text{ kg} \cdot \text{m}^2$ $\sum J = 1.15m(J_1 + J_2) + \frac{(Q + TL)D_0^2 \eta}{4i^2}$

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起动时间: $t_q = 0.7$ s
Starting time:

允许的起动时间: $[t_q] = 4.0$ s
Allowable starting time:

需延长起动时间
It needs to extend the starting time

② 起动平均加速度

② Average Starting Acceleration

平均加速度: $a_q = 0.57$ m/s²
Average acceleration:

允许的起动平均加速度: $[a_q] = 0.10$ m/s²
Allowable average starting acceleration:

起动转矩不可过大, 否则会产生较大冲击!
The starting torque shall not be too large otherwise it may lead to large impact!

7. 缓冲器

7. Buffer

(1) 选型

(1) Model Selection

$$W = \frac{m_{hc} v_c^2}{2} - P_{mt} S = E_t - P_{mt} S = \frac{1}{2} c S^2 \times n$$

最大减加速度: $a_{\max} = 4.0$ m/s²
Maximum deceleration and acceleration:

运行阻力: $P_{mt} = 1.60$ kN
Travelling resistance:

碰撞瞬时动能: $E_t = 0.4$ kN·m
Collision instantaneous kinetic energy:
 $\Delta = 3738.4$

缓冲行程: $S = 43$ mm
Buffer stroke:

碰撞质量: $m_{hc} = 4.90$ t
Collision quality:

碰撞前的速度折减系数;
Velocity reduction coefficient before collision: 100%

碰撞速度: $v_c = 0.42$ m/s
Collision speed:

单个缓冲器吸收的能量: $W = 0.2$ kJ
Energy absorbed by single buffer:

小车缓冲器型号: **HCPU60-A** 聚氨酯缓冲器
Buffer model of hoist trolley: **HCPU60-A** PU buffer

缓冲器容量: $[W] = 2.030$ kJ
Buffer capacity:

容量满足要求!
The capacity meets the requirements!

最大撞击力: $P_e = 148.0$ kN

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Maximum impact force:

缓冲器许用行程:
Allowable buffer stroke: $[S] = 43$ mm

实际行程为缓冲器的许用行程!
The actual stroke is the allowable buffer stroke!

弹性系数:
Elastic coefficient: $c = 2.1958$ kN/mm

(弹性缓冲器)
(Elastic buffer)

缓冲器效率:
Buffer coefficient: $\eta_h = 0.85$

单边个数:
Number of single edge: $n = 2.0$

单个重量:
Single weight: $m_{31} =$ kg

(2) 单个缓冲器末端力校核
(2) Single Buffer End Force Check

缓冲器的末端力:
End force of buffer:

$$P_{\max} = \frac{cS}{\eta} = 111.08 \text{ kN}$$

缓冲力满足 要求!
The buffer force meets the requirements!

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